Assignment 3

due date: June 20, 2024

1. Let $\{B(t), t \geq 0\}$ be a standard Brownian motion. For $s \leq t$, determine the distribution of $B(s) + B(t)$.

2. For $t \geq 0$, define $X(t) = Z\sqrt{t}$ where $Z \sim N(0, 1)$.
(a) What is the distribution of $X(t)$?
(b) Is $\{X(t), t \geq 0\}$ a Brownian motion? Why?

3. Let $\{W(t), t \geq 0\}$ and $\{B(t), t \geq 0\}$ be two independent standard Brownian motions. Let ρ be a constant such that $-1 < \rho < 1$. Define the stochastic process $\{X(t), t \geq 0\}$ as
$$X(t) = \rho W(t) + \sqrt{1 - \rho^2} B(t)$$
(a) Show that $\{X(t), t \geq 0\}$ is a standard Brownian motion.
(b) Determine the covariance and correlation coefficient of $X(t)$ and $W(t)$.

4. Let $\{B(t), t \geq 0\}$ be a standard Brownian motion.
(a) Determine $P(B(2.1) > 1.7 | B(1.6) = 0.5)$.
(b) Determine $P(B(1.6) \leq 0.5 | B(2.1) = 1.7)$.
(c) Show that for any $0 < s < t$, $B(t)$ and $B(s) - \frac{s}{t} B(t)$ are independent.

5. Let $\{B(t), t \geq 0\}$ be a standard Brownian motion and $0 < s < 1$.
(a) What is the distribution of $B(s) + B(s^2)$?
(b) What is the distribution of $B(s) + B(s^2) + B(1)$?
(c) What is the distribution of $B(s)$ given that $B(1) = 0$?
(d) Determine $E(B(s)B(s^2) | B(1) = 0)$.

6. Let $\{B(t), t \geq 0\}$ be a standard Brownian motion.
(a). Let $Z(t) = |B(t)|$. Determine the probability density function of $Z(t)$.
(b). Let T_a be the first time $B(t)$ hits a, where $a > 0$, and $Y(t)$ be the process absorbed at a, i.e.
$$Y(t) = \begin{cases} B(t) & \text{if } t < T_a \\ a & \text{if } t \geq T_a \end{cases}.$$
Determine the cumulative distribution function of $Y(t)$.